

# ITT320 INTRO TO COMPUTER SECURITY

## Chapter 8 : Defending Against Virus Attacks

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# Objectives

- Explain how virus attacks work
- Explain how viruses spread and how they propagate
- Distinguish between different types of virus attacks
- Employ virus scanners to detect viruses
- Have a working knowledge of several specific viruses
- Formulate an appropriate strategy to defend against virus attacks



# Introduction

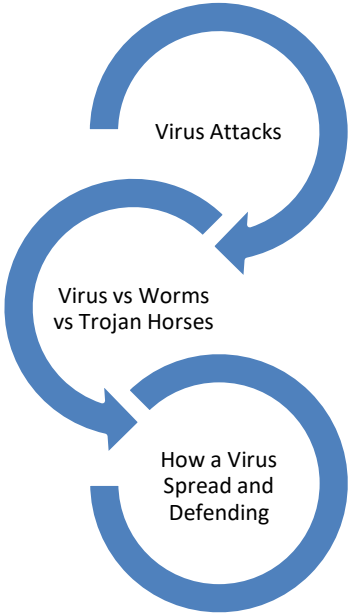
Defending against virus attacks is more than having anti-virus software in place.

Organizations need to understand the nature of viruses in relation to other types of attacks so that they can reduce the overall effects and impact that viruses will have on their organization

# Understanding Virus Attacks

- Important points that should be answered to help understand Virus Attacks :

  - What is a Computer Virus?
  - What is a Computer Worm?
  - How does a Virus and Worm Spread?



# RECAP
# Classifications of Threats

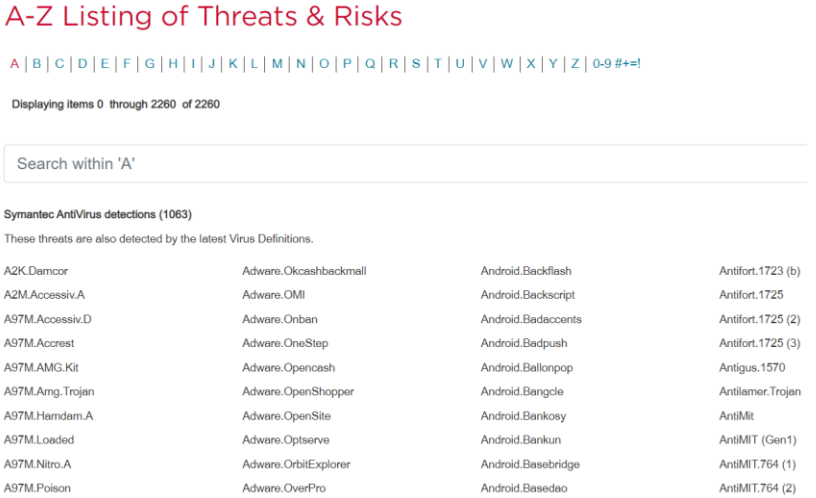
Malware Attack – Stand for Malicious Software, the most prevalent danger to your network

Example of Malware:

- ✓ **Virus**
- ✓ **Worms**
- ✓ **Trojan Horses**
- ✓ **Adware**
- ✓ **Spyware**
- ✓ **Bot**




*Ransomware (securityintelligence.com)*




*https://www.broadcom.com/support/security-center/a-z*

# Computer Virus

- A computer virus is a type of malicious code or program / software written to alter/harm the way computer operates and when executed, replicate itself and spread to other machines.

  - Self-Replicates
  - Spreads Rapidly
  - May or may not have a malicious payload



# How does a Virus Spread?

## How a Virus Spreads:

- Finds a network connection; copies itself to other hosts on the network
  - Requires programming skill
- Mails itself to everyone in host's address book
  - Requires less programming skill



*Image : CyberThreatPortal.com*

# How does a Virus Spread?

## E-mail Propagation

- More common for one major reason;
  - Microsoft Outlook is easy to work with
  - Five lines of code can cause Outlook to send e-mails covertly

Other viruses spread using their own e-mail engine.



*Image : CyberThreatPortal.com*

# How does a Virus Spread?

Network Propagation

- Less frequent, but just as effective

Website Delivery

- Relies on end-user negligence

Multiple vectors for a virus are becoming more common



*Image : CyberThreatPortal.com*

# Example of Viruses

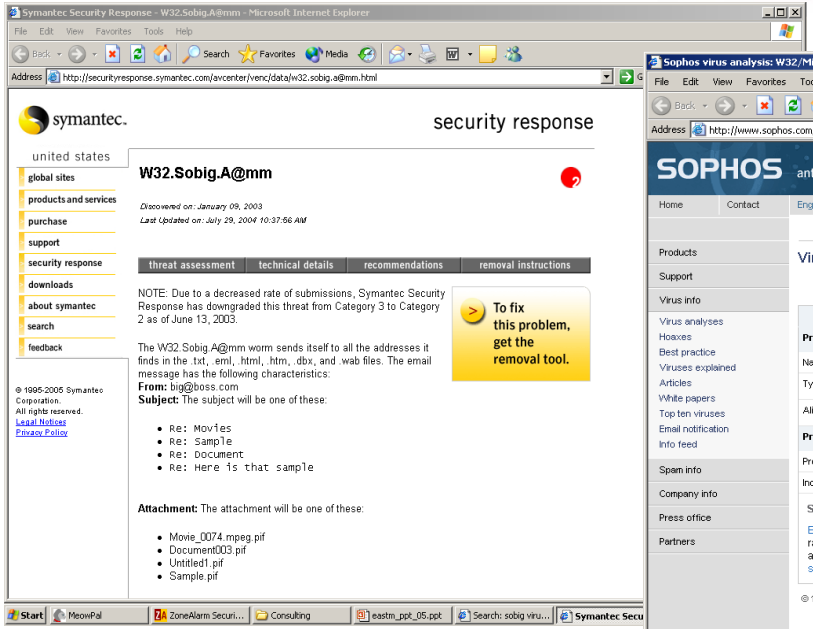
- Virus Types
  - Macro
  - Multi-Partite
  - Armored
  - Memory Resident
  - Sparse Infector
  - Polymorphic




*Realtime and Latest : https://threats.kaspersky.com/*

# Example of Viruses

- Examples
  - Rombertik
  - Gameover ZeuS
  - FakeAV




Sobig Virus

Minmail Virus



# Virus Scanner

Software that tries to prevent viruses from infecting machines.

Work in two ways, generally:

1. Contains a list of known virus files in a .dat file and compares files on your computer to that file
2. Monitor the computer for certain types of virus behavior




Can be on-demand or ongoing scanning

# Virus Scanning Techniques

In general, **Antivirus** software scans a file, program, or an application and compares a specific set of code with information stored in its database.

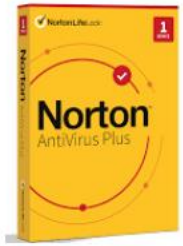



1. E-mail and attachment scanning
2. Download scanning
3. File scanning
4. Heuristic scanning
5. Active code scanning
6. Instant messaging scanning

# Commercial Antivirus Software

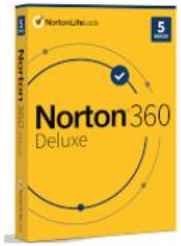
Factors to consider when choosing antivirus:

- Budget (Price)
- Vulnerability (how often is e-mail used or files downloaded?)
- Skill (users need to understand how to use it)
- Technical (specifications of the software and how it functions)



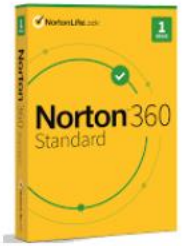

Norton Antivirus Plus 2021

**MYR 59.00**




Norton 360 Deluxe - 5 Devices - MYR 8...

**MYR 129.00**




Norton 360 Antivirus 2021 for ...

**MYR 79.00**




80% Off - Windows Antivirus 2020,...

**MYR 77.06**




80% Off - Internet Security (Total AV...

**MYR 77.06**




Norton 360 Premium with...

**MYR 219.00**



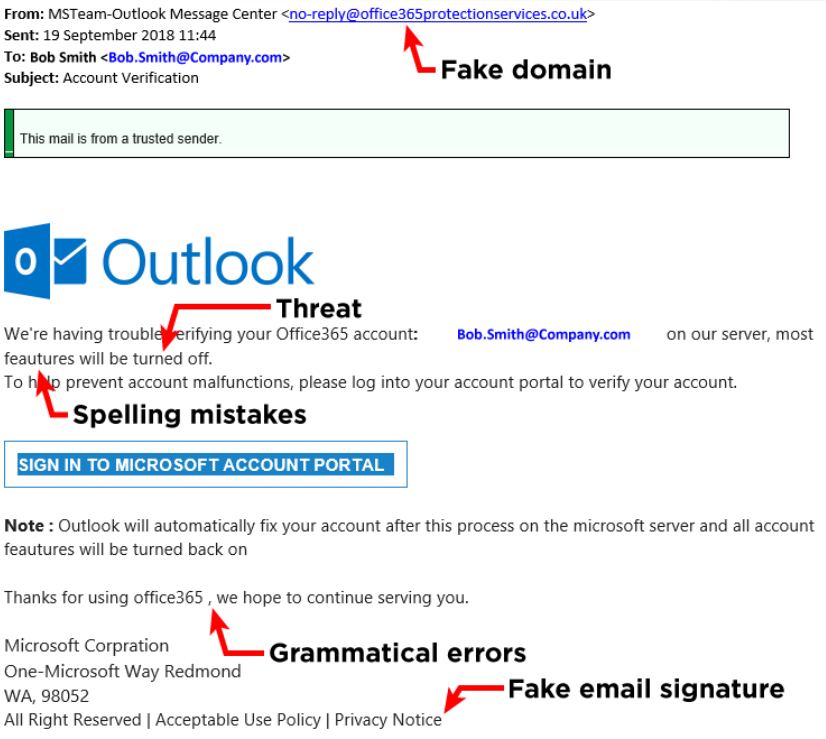

McAfee: Total Protection 2021 (...

**MYR 98.71**

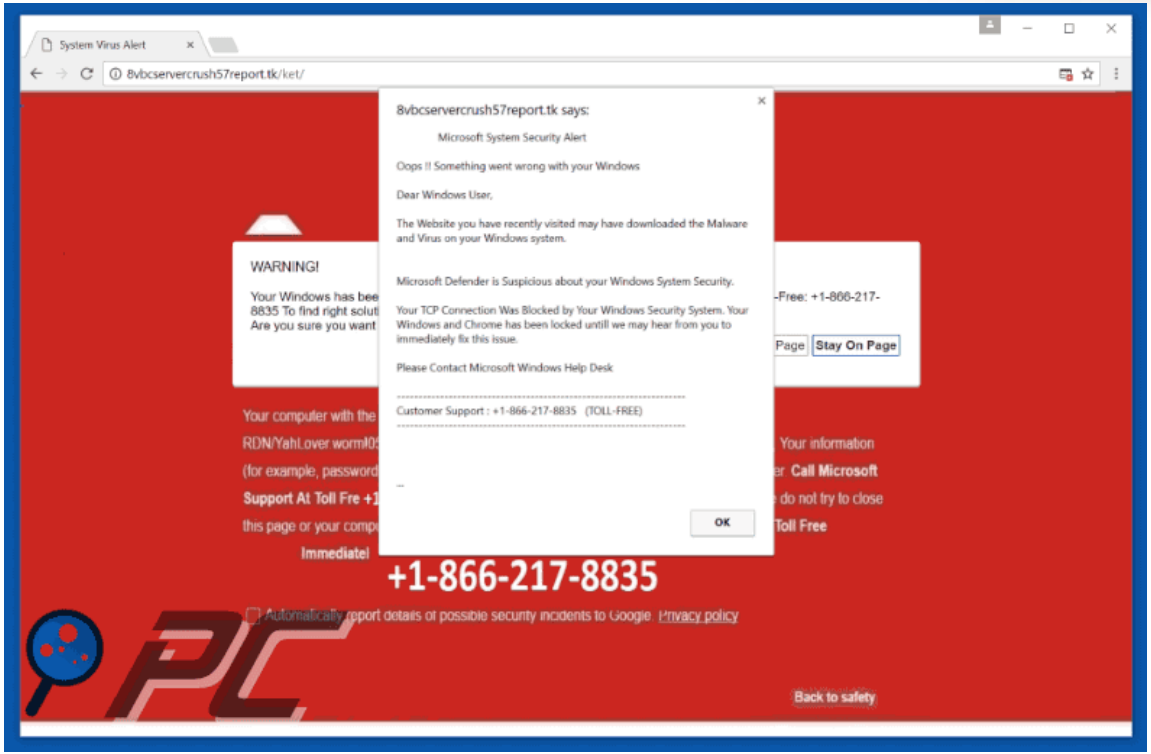
# Antivirus Policies and Procedures

## Brief summary of possible policies:

- Always use a virus scanner
- If you are not sure about an attachment, do not open it
- Consider exchanging a code word with friends
- Do not believe “security alerts” you are sent
- Be skeptical of any e-mail you are sent
- Do not download files from the Internet



# Antivirus Policies and Procedures




FAKE SECURITY ALERT!

# Defending Your System

Additional Methods for Defending Your System :

- Set all browsers to block active code
- Set all user accounts so that they cannot install software or change browser security
- Segregate subnetworks



# Infected by a Virus?

**What to do if your system is infected by a Virus ?**

Need to focus on ‘THREE’ STEPS :

1. Stopping the Spread of the Virus
2. Removing the Virus
3. Finding out how the infection Started



# Infected by a Virus?

**1. Stop the Spread of the Virus**

Priority / Must Do in First Place. Follow the Steps Below :

1. If infection is on a WAN, disconnect the WAN
2. If on a subnet, disconnect that subnet
3. Disconnect vital servers that might be connected to an infected machine
4. Disconnect any backup devices that might be connected to an infected machine.



# Infected by a Virus?

**2. Removing Virus**

Virus Propagation Must be Stopped First

Run Antivirus:

Update Your AV | Run Full Scanning

Find Removal Instructions on The Internet

✓ Some Viruses Cannot be Removed



# Infected by a Virus?

**3. Find Out How the Infection Started**

Talk to users of infected machines

- Read any online documentation on that virus.
- Check activity logs from the machine.





# SUMMARY

- Virus attacks and hoaxes are arguably the greatest threat to computer networks
- Sophistication of viruses and worms is increasing
- It is necessary to understand how viruses work in order to prevent infection
- One also needs to know how viruses spread



# SUMMARY

- There are a number of ways to reduce exposure to viruses

Virus scanners :

- Understand how they work
- Be familiar enough to choose the right one for your organization
- Come in both commercial and free versions

- Establish written policies and procedures



# SUMMARY

- There are a number of ways to reduce exposure to viruses (continued)m

  - Block installation of software by users
  - Secure the browser
  - Separate subnetworks

- Security should have a multi-layer approach